

BASF
We create chemistry

Care Creations™

Replexium®

Achieve your beauty aspiration

by Beauty Creations
another Care Creations™ product group Inspired by Life

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SUMMARY FILE

Replexium®	**BC10091**
Origin - Description	Skin-bioavailable synergistic complex of 2 peptides
Regulatory data	
INCI	Dimethyl Isosorbide (and) Polysorbate 20 (and) Water (and) Acetyl Tetrapeptide-11 (and) Acetyl Tetrapeptide-9
China	All components are listed in Inventory of Existing Cosmetic Ingredient in China (IECIC 2015)
CAS#	5306-85-4; 9005-64-5; 7732-18-5; 928006-88-6; 928006-50-2
EINECS#	226-159-8; 500-018-3; 231-791-2; /; /
Appearance	Syrupy liquid from pale yellow to yellow, with characteristic odor
Preservative	None
Natural labels	None
Cosmetic use	
Properties	• Stimulation of key epidermal and dermal components • Increases proteoglycans lumican and syndecan-1 • Increases collagen XVII • Contributes to functional organization of the extracellular matrix • Contributes to cell-cell and cell-matrix adhesion in dermis, epidermis and dermal-epidermal junction • Anti-wrinkle • Increase of dermis density and skin firmness
Applications	• Performance skin care • Anti-aging skin care • Skin firming • Youth-keeper lotion
Formulation data	
Concentration of use	2%
Solubility	Soluble in water, insoluble in oils and fats
Incorporation method	Replexium® is incorporated during the final process below 40°C, or at room temperature for cold processing
Optimal pH	4-8
Patent family	Application filed



CONSUMER'S BEAUTY ASPIRATIONS

The world population is aging. Life expectancy is increasing worldwide and the aged fraction of the population is the fastest growing [World Health Organization, 2013].

In a consumer survey conducted by BASF, we asked 235 women, ages 40-73 years, about their aspired age.

When asked to picture themselves in their mind, and how many years younger do they look or want to look, 60% wished they looked 5-10 years younger than their actual age.

When asked "How many years younger do you believe a high quality skin care product can make you look?", 80% expect 5-10 years younger looking appearance.

A majority of consumers wish to look between 5-10 years younger than their actual age and expect a quality skin-care product would help them achieve their aspirations.

CONSUMER SURVEY

235 women between 40 to 73 years old participated to this survey in North America. Respondents received a link to the survey via email and completed the questionnaire at home. They were asked to enter a number to the following questions:

"Close your eyes. Picture yourself in your mind's eye. Think about your face and how you look. In this mental image of yourself, how many years younger do you look or want to look?"

"How many years younger do you believe a high quality skin care product can make you look?"

IMPROVING SKIN AGING, BY INDUCING CHANGES IN EXTRACELLULAR MATRIX, EPIDERMAL COHESION AND DERMAL-EPIDERMAL JUNCTION STRUCTURAL INTEGRITY

Women wish they look younger than their actual age. To improve the signs of skin aging, it is important to understand what makes that the skin looks older.

Age-dependent changes affect the skin's structure and functions in the dermis, the epidermis, and the dermal-epidermal junction (DEJ) leading to a loss of firmness, wrinkles and an increased skin fragility.

With aging, there is a progressive atrophy of the dermis with changes to the dense collagen-rich extracellular matrix (ECM) produced by fibroblasts. Collagen I fibrils are the primary structural component of this connective tissue conferring strength and mechanical support to the skin. During aging, intact tightly packed and well-organized collagen I fibrils are fragmented and disorganized due to altered fibrils homeostasis, leading to impaired structural integrity and mechanical properties of the skin [Quan and Fisher, 2015].

The epidermis, which is the first barrier protecting the body from the external environment and dryness, is also affected by aging with a reduced mitosis of the epidermal layer and a loss of cohesion of cells [Neerkens *et al*, 2004].

The epidermis is anchored to the dermis by a complex network forming the dermal-epidermal junction (DEJ). The DEJ is also impacted by aging with a flattening and a less effective anchoring system, resulting in a reduced surface area and a substantial increased skin fragility [Hatje *et al*, 2015].

Lumican and syndecan-1 proteoglycans: dermal and epidermal targets to fight against skin aging

Proteoglycans (PGs) conjugate a core glycoprotein with one or several covalently attached sulfated glycosaminoglycans (GAGs) chains (Figure 1).

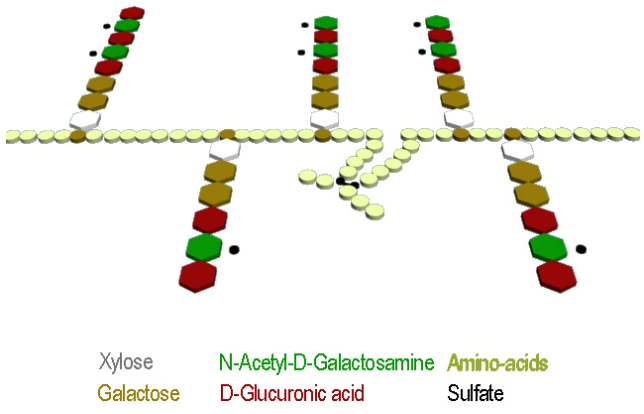


Figure 1 - General structure of proteoglycans.

Proteoglycans are present both in the epidermis and the dermis and are involved in various structural functions and physiological regulations. They are ubiquitous macromolecules of extracellular matrices (Figure 2) and of cell membranes and act as biomechanical supports, tissue organizers and biological filters contributing to tissue hydration, resistance and resilience, cell-cell and cell-matrix interaction. Due to their capacity to interact with positively charged proteins through their polyanionic sulfated carbohydrate chains, they also constitute the biological reservoir of many cytokines and growth factors [Maquart et al, 2010].

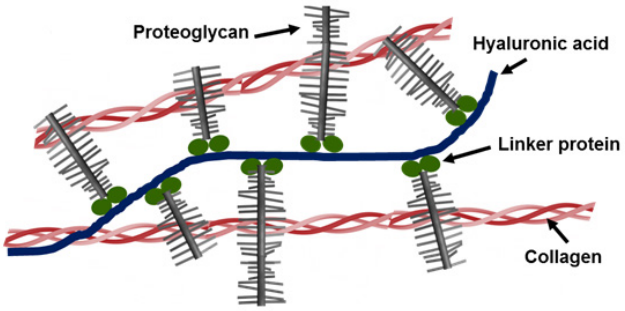


Figure 2 - Schematic representation of proteoglycans in the extracellular matrix .

Lumican, a proteoglycan of the dermal extracellular matrix

Lumican belongs to the family of Small Leucine-Rich Proteoglycans (SLRPs) which includes also decorin and biglycan, present in many tissues, and characterized by their relatively low molecular weight and leucine-rich repeats.

Lumican was first identified as a major proteoglycan of the cornea [Funderburgh and Conrad, 1990]. In adult human skin, it is also present in the form of a glycoprotein [Grover *et al*, 1995]. It is located in the dermis where it is involved in the formation and stabilization of collagen I fibrils (Figure 3), and consequently in the extracellular matrix organization and skin resistance [Chakravarti *et al*, 1998]. Lumican increases the speed of fibril formation and enhances collagen I fibril stability [Neame *et al*, 2000]. Recent studies have also shown that lumican plays a pivotal role in skin wound healing by facilitating collagen fibrillogenesis [Yeh *et al*, 2010] and the activation and contraction of fibroblast in the wound [Liu *et al*, 2013].

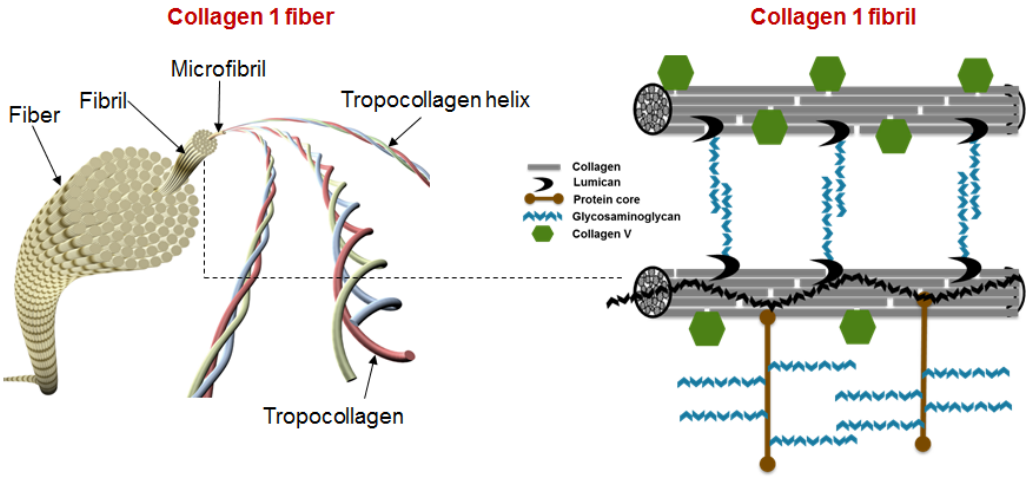


Figure 3 - Schematic representation of the role of lumican in Collagen 1 fibrils organisation in the dermis.

BCS Research: Lumican decreases with age

The expression of lumican, decorin and biglycan (SLRPs) was studied on cultured fibroblasts obtained from young to old donors [Vuillermoz *et al*, 2005].

Using a Northern Blot analysis, this study showed a significant decrease ($p<0.01$) of lumican mRNA expression with the donor's age (Figure 4). By contrast, no correlation was found for either decorin or biglycan mRNA.

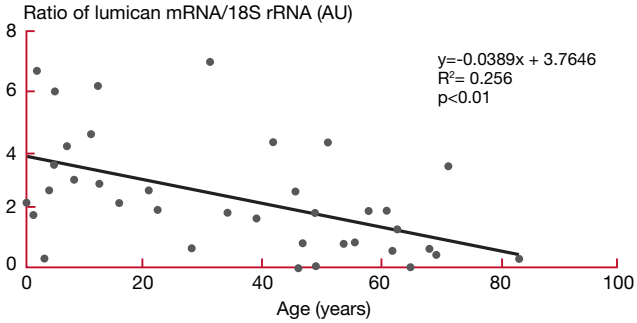
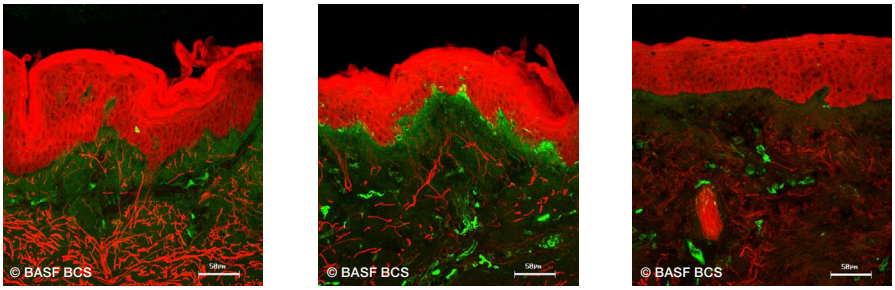


Figure 4 - Lumican mRNA expression in fibroblasts as a function of the donor's age (Northern Blot). Lumican mRNA level normalized to 18S rRNA.

Using immunohistochemistry on the biopsies, the changes of lumican protein with age was also analyzed. Lumican was particularly located in the superficial papillary layer of the dermis, and was significantly decreased in aged skin (donors over 50 years old) compared to young (0-15 years old) and adult (16-50 years old) skin, respectively by -81% ($p<0.01$) and -85% ($p<0.01$) (Figure 5A and 5B).

A



B

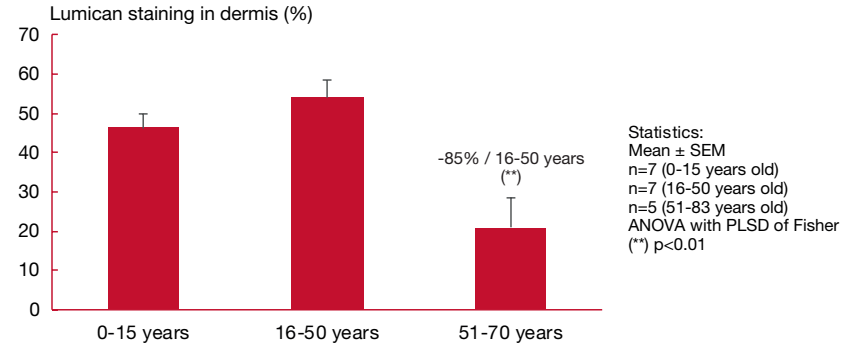


Figure 5 - Variation of lumican protein expression in the dermis as a function of donor's age. **A**: visualization of skin sections (IHC, lumican labelled in green). Scale bar: 50 µm. **B**: Quantification in aged skin (donors over 50 years old) compared to young (0-15 years old) and adult (16-50 years old) skin (IHC and image analysis).

From these results, it may be suggested that progressive collagen I fibril denaturation with aging is linked to a defect in lumican expression with an impaired collagen I fibril organization. By maintaining the narrow diameter of fibrils and decreasing interfibrillar space, stimulation of lumican synthesis should improve collagen I fibril architecture and stability in the extracellular matrix, and overall tissue biomechanics.

Syndecan-1, an epidermal cell membrane proteoglycan

Syndecan-1 is a proteoglycan that belongs to the syndecan family, and is the most abundant syndecan in epithelial keratinocytes. It is mainly expressed in the suprabasal layer of the epidermis [Sanderson et al, 1992], and markedly upregulated in all cell layers of the epidermis during wound healing [Elenius et al, 1991]. Syndecan-1 is an important modulator of epidermal cell proliferation, migration and adhesion [Ojeh et al, 2008]. It is involved in complex signaling as a receptor of growth factors and cytokines (e.g. FGFs, VEGF, TGFβ), as a receptor of extracellular matrix macromolecules (integrins), and as a co-receptor of tyrosine kinase receptors (e.g. VEGFR, FGFR, EGFR) [Carey et al, 1997; Maquart et al, 2010; Chung et al, 2016]. It plays a crucial role in the epidermal cohesion (Figure 6) and skin regeneration.

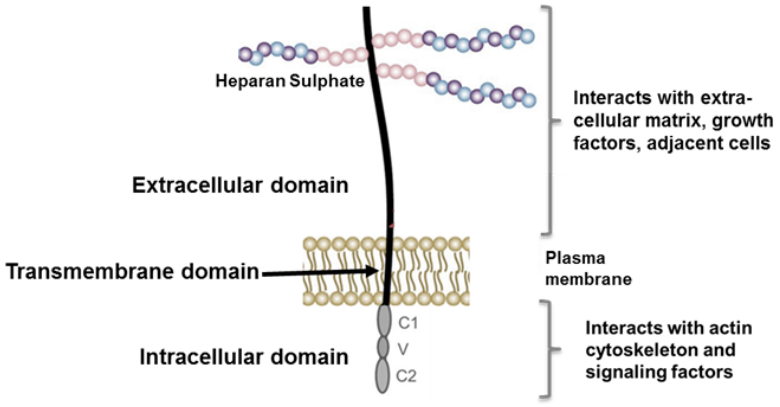


Figure 6 - Schematic representation of the role of syndecan-1 in the epidermal cohesion.

BCS Research: Syndecan-1 decreases with age

The expression of syndecan-1 was studied on cultured keratinocytes obtained from young to old donors [Wegrowski et al, 2005]. Using Northern Blot analysis, this study showed a decrease of syndecan-1 mRNA expression with donor's age (Figure 7).

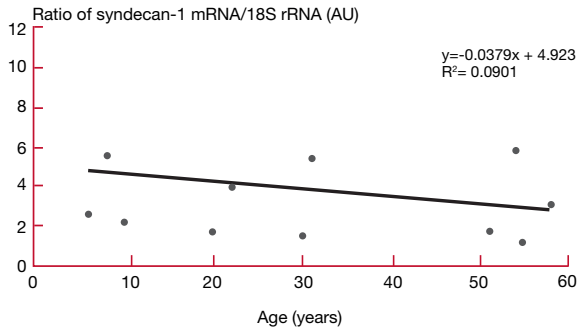
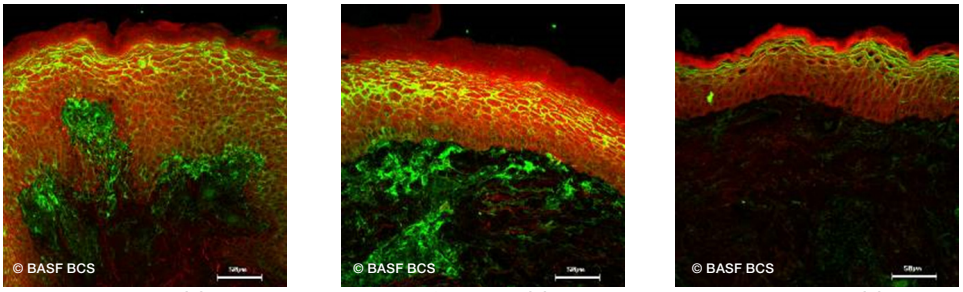


Figure 7 - Lumican mRNA expression in fibroblasts as a function of the donor's age (Northern Blot). Lumican mRNA level normalized to 18S rRNA.

Using immunohistochemistry on the biopsies, the changes of syndecan-1 protein with age was also analysed. On these biopsies, Syndecan-1 was predominantly located around keratinocytes predominantly in the epidermal supra-basal layers. Some staining of vessel cells and fibroblasts was also observed in the dermis. Syndecan-1 expression was significantly decreased in aged skin (donors over 50 years old) compared to young (0-15 years old) and adult (16-50 years old) skin, respectively by -55% ($p<0.05$) and -62% ($p<0.01$) (Figure 8A and 8B).

A



B

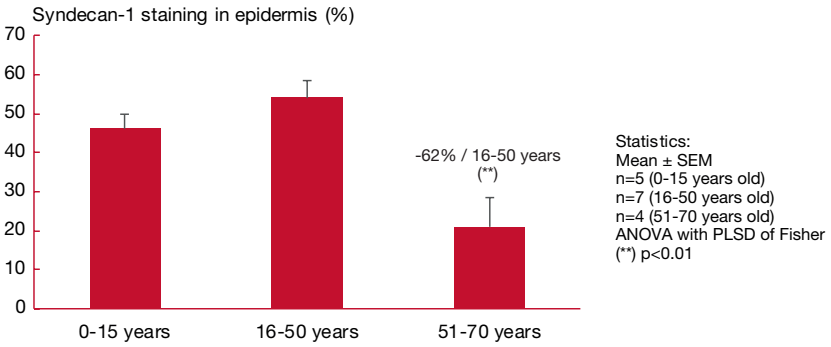


Figure 8 - Variation of syndecan-1 protein expression in the epidermis as a function of donor's age. **A**: visualization of skin sections (IHC, syndecan-1 labelled in green). Scale bar: 50 µm. **B**: Quantification in aged skin (donors over 50 years old) compared to young (0-15 years old) and adult (16-50 years old) skin (IHC and image analysis).

Similar results have also been reported by Oh *et al* in 2011 from a study on skin biopsies, showing that syndecan-1 protein expression was decreased with intrinsic skin aging in both genders [Oh *et al,* 2011].

From these results, it may be suggested that the decreased proliferation potential and cohesion of keratinocytes in an aging epidermis is linked to a defect in syndecan-1 expression with an alteration of cell signalling, proliferation, cell-cell and cell-matrix adhesion.

Collagen XVII: a DEJ target to fight against skin aging

Collagen XVII, a DEJ anchoring transmembrane protein

The dermal-epidermal junction (DEJ) is a basement membrane that joins and maintains the association between the cellular epidermis and connective-tissue-rich dermis. It is composed of a complex network of macromolecules interconnecting its different components to provide the skin with both structural integrity and mechanical stability.
Collagen XVII is a collagenous transmembrane protein and a structural component of the DEJ anchoring complex [Powell *et al*, 2005] (Figure 9). It is a component of the hemidesmosomes, which mediate the attachment of the basal keratinocytes of the basal keratinocytes of the epidermis to the underlying basement membrane.

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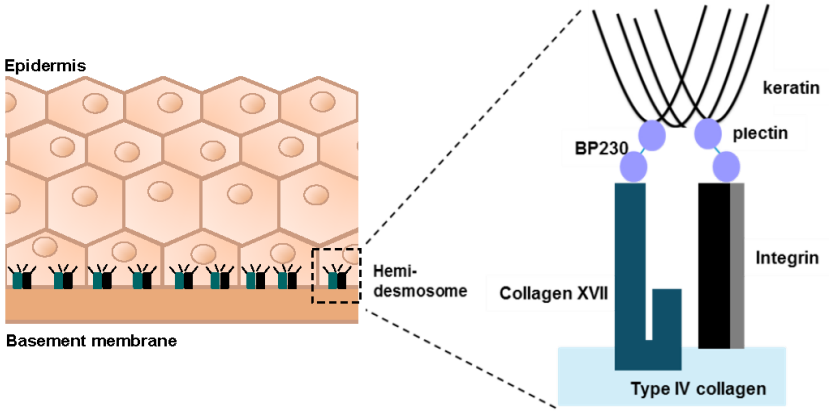


Figure 9 - Schematic representation of the role of collagen XVII in the dermal-epidermal junction.

Studies have shown that collagen XVII has a key role in the assembly and functioning of this cell-matrix adhesion structure through the stabilization of the hemidesmosome complex. Mutations in the collagen XVII gene (COL17A1) led to a decrease of epidermal adhesion, as well as skin fragility (blistering) in response to minimal mechanical deformation [Powell *et al*, 2005]. In addition, collagen XVII is involved in transmembrane signal transduction and in the regulation of keratinocyte differentiation [Zillikens and Guidice, 1999].

Collagen XVII decreases with age

The changes in collagen XVII proteins with aging was reported recently on photo-protected buttock skin biopsies obtained from donors 18 to 75 years old [Langton *et al*, 2016] and are illustrated in Figure 10.

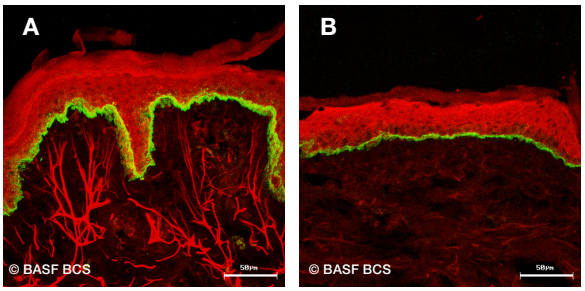


Figure 10 - Visualization of collagen XVII in the DEJ (IHC, collagen XVII labeled in green). Scale bar: 50 µm. **A** – Typical young skin. **B** – Typical old skin.

Collagen XVII expression was significantly decreased in the skin from aged (65-75 years old) compared to young donors (18-30 years old). Skin of the aged participants concomitantly exhibited a characteristic flattened DEJ with rete ridge shortening.

From these results, it may be suggested that the loss of structural integrity of the aging DEJ and increased fragility is linked to this reduced level of collagen XVII at the DEJ.

CHALLENGES AND STRATEGIES

Women aspire to look younger than their actual age, and expect a quality skin-care product would provide a non-invasive alternative to aesthetical surgery and help them to achieve their beauty aspiration.

Peptides offer the solution as effective ingredients for the development of **performance skin care products** meeting the consumer's expectation.

Replexium® combines 2 patented peptides with bioavailability agents as a multipronged approach to address **skin aging**: to improve extracellular matrix, epidermal cohesion and dermal-epidermal junction structural integrity.

It targets (Figure 11) the improvement of:
- collagen fibrils in the extracellular matrix by stimulating the dermal proteoglycan **lumican**,
- cohesion and proliferation of epidermal cells by stimulating the epidermal proteoglycan **syndecan-1**,
- adhesion of basal keratinocytes to DEJ by stimulating **collagen XVII**,

for quick **wrinkle** appearance reduction and skin **firming** benefits in only a few weeks.

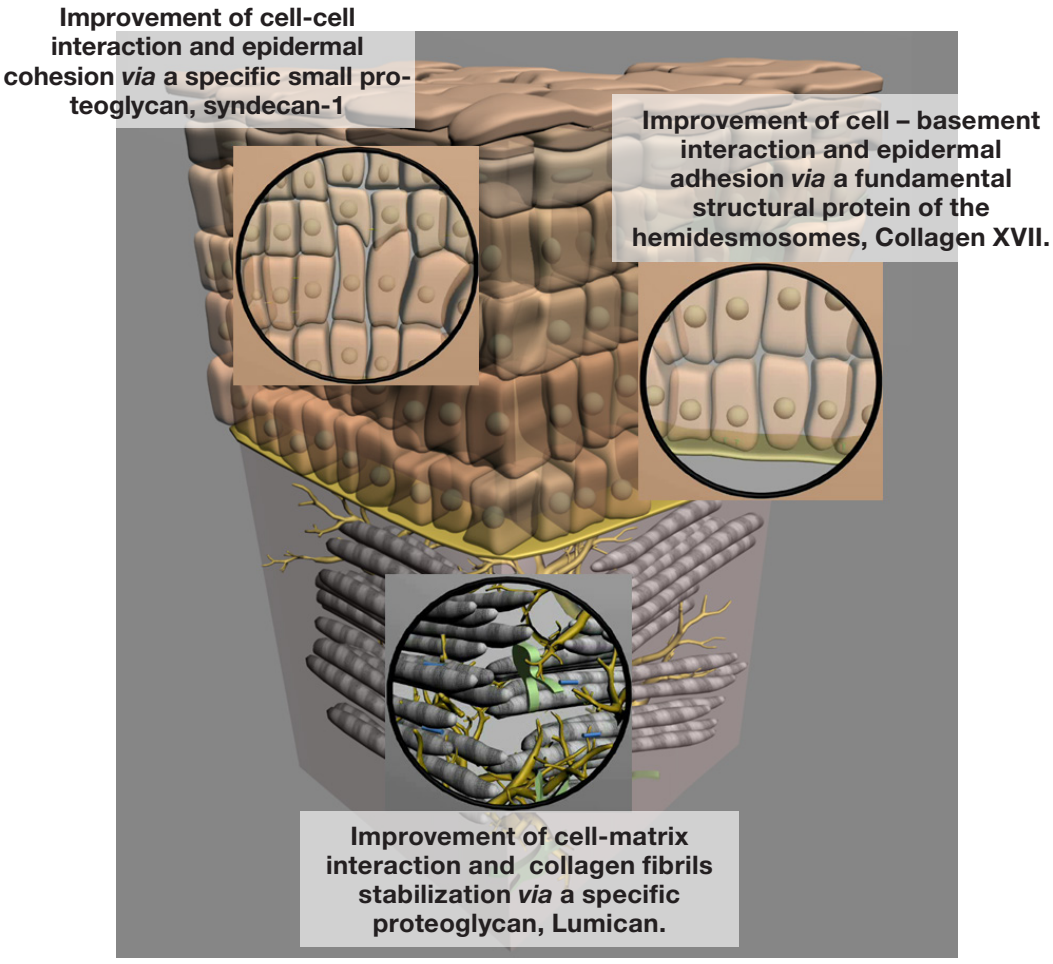
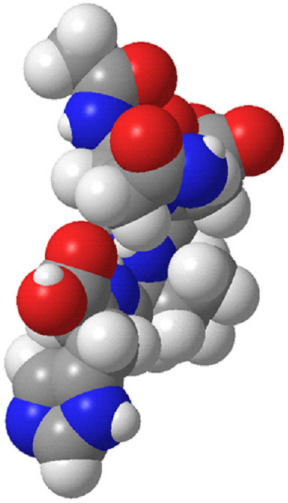


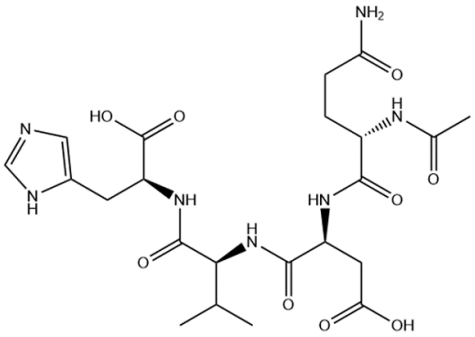
Figure 10 - Schematic representation of the skin targets of the peptide complex Replexium®.

REPLEXIUM®, A SYNERGISTIC COMPLEX OF 2 PEPTIDES



Acetyl Tetrapeptide-9

Sequence: N-Acetyl-L-Glutaminyl-L-Aspartyl-L-Valyl-L-Histidine



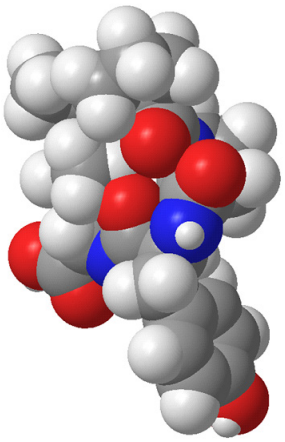
Synonyms: N-Acetyl-Gln-Asp-Val-His-OH; Ac-QDVH

CAS RN: 928006-50-2

Molecular weight: 539.5 g/mol

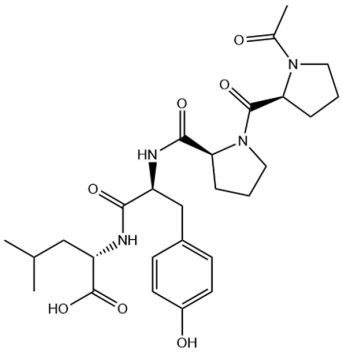
Process: chemical peptide synthesis, followed by N-acetylation. All amino-acids are in the natural L-form and not of animal origin.

Peptide sequence: QDVH sequence is identical to a fragment of the human PTH which activates the PKC-pathway [Wu and Kumar, 2000].



Acetyl Tetrapeptide-11

Sequence: N-Acetyl-L-Prolyl-L-Prolyl-L-Tyrosyl-L-Leucine



Synonyms: N-Acetyl-Pro-Pro-Tyr-Leu-OH; Ac-PPYL

CAS RN: 928006-88-6

Molecular weight: 530.6 g/mol

Process: chemical peptide synthesis, followed by N-acetylation. All amino-acids are in the natural L-form and not of animal origin.

Peptide sequence: PPYL sequence is identical to a fragment of a mammalian antibacterial proline-rich cathelicidin polypeptide PR-39 expressed in wound, which was reported to stimulate the expression of syndecan [Chan *et al*, 2001].

Safety and tolerability data of the product

Replexium® was evaluated to ensure its safety in the recommended conditions of use. Replexium® does not irritate the eyes or skin and no indication of skin sensitization was observed.

Description

Replexium® (ref. BC10091) is a preservative free, water-soluble solution of 350-650 ppm of Acetyl Tetrapeptide-9 and 525-825 ppm of Acetyl Tetrapeptide-11 with skin-bioavailability agents.

INCI name: Dimethyl Isosorbide (and) Polysorbate 20 (and) Water (and) Acetyl Tetrapeptide-11 (and) Acetyl Tetrapeptide-9

Dose of use: 2%

Replexium® is also available on demand as a powder without bioavailability agents under the reference Replexium® PW BC10082 (INCI name: Mannitol (and) Acetyl Tetapeptide-11 (and) Acetyl Tetrapeptide-9; dose of use 1%).



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Example: Illustration of a commercial sample of Replexium® incorporated into a serum and a face cream at 2%.



DEMONSTRATED ANTI-AGING EFFECTIVENESS

The ability of Replexium® to fight against skin aging was proven *in vitro* by the stimulation of age-affected cellular mechanisms leading to a structurally weakened skin, and *in vivo* on the improvement of signs of skin aging on the face.

In vitro, we demonstrated the effect of:

- the Acetyl Tetrapeptide-9 component of Replexium® on the stimulation of lumican synthesis,
- the Acetyl Tetrapeptide-9 component of Replexium® on the stimulation of human dermal fibroblasts cell number and collagen I synthesis,
- the Acetyl Tetrapeptide-11 component of Replexium® on the stimulation of syndecan-1 synthesis,
- the Acetyl Tetrapeptide-11 component of Replexium® on the stimulation of human epidermal keratinocytes proliferation,
- the Acetyl Tetrapeptide-11 component of Replexium® on the stimulation of collagen XVII synthesis,
- the complex of the two peptides on the density and organization of the extracellular collagen fiber network in 3D bioprinted skin.

In vivo, we demonstrated the anti-aging properties of Replexium® on dermal density and wrinkles (forehead wrinkles, crow's feet, and nasolabial folds) compared to a leading peptide-blend benchmark in a clinical study.

EFFICACY

Stimulation of lumican synthesis by Acetyl Tetrapeptide-9

in vitro

OBJECTIVE

Lumican proteoglycan is involved in the collagen I fibril organization for a dense, regular and functional extracellular matrix, but decreases with aging.

The aim of the study was to evaluate the stimulatory effect of Acetyl Tetrapeptide-9 on lumican synthesis by senescent human dermal fibroblasts in two *in vitro* models: monolayer cell culture and pseudo-dermis (fibroblasts embedded in collagen lattices).

In vitro aging of fibroblasts was induced by several consecutive culture passages (>15) of human primary fibroblasts according to Hayflick's model [Passi *et al*, 1997].

RESULTS & DISCUSSION

***In vitro* on monolayer fibroblast culture**

Interleukin-4, a positive reference, was used to validate the model, with measurement of lumican by immunohistochemistry and image analysis (74% increase vs control, $p<0.05$, data not shown).

Acetyl Tetrapeptide-9 showed a dose-dependent significant stimulation of lumican synthesis by aged human fibroblasts (Figure 12A and 12B), up to 115% increase vs control at 2.2 µg/ml ($p<0.001$).

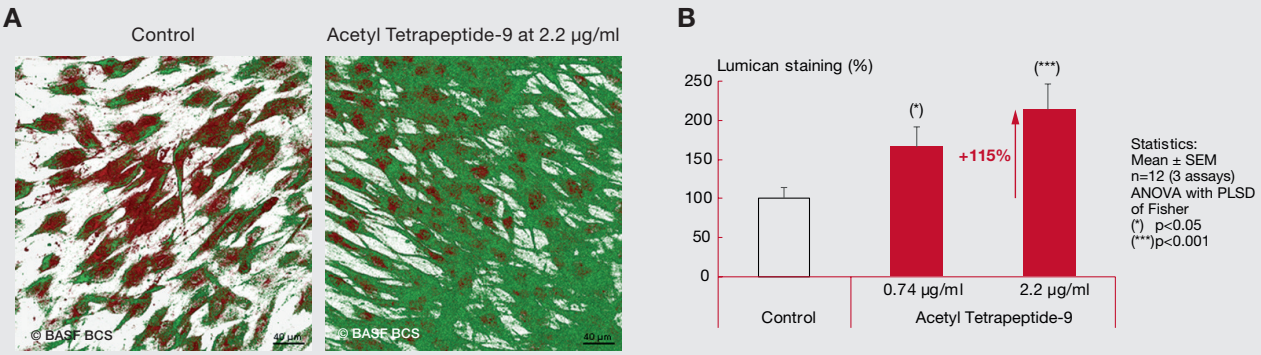


Figure 12 - Stimulation of lumican synthesis by Acetyl Tetrapeptide-9 in aged human dermal fibroblasts in monolayer culture. **A**: Visualization (ICC, lumican labelled in green). Scale bar: 40 µm. **B**. Quantification of staining as number of stained pixels x fluorescence intensity in green channel, expressed as percentage of untreated control (image analysis).

***In vitro* on pseudo-dermis (fibroblasts embedded in collagen lattices)**

Acetyl Tetrapeptide-9 at 2.2 µg/ml significantly stimulated by 58% ($p<0.05$) lumican synthesis by aged human fibroblasts embedded in collagen lattices (figure 13A and 13B).

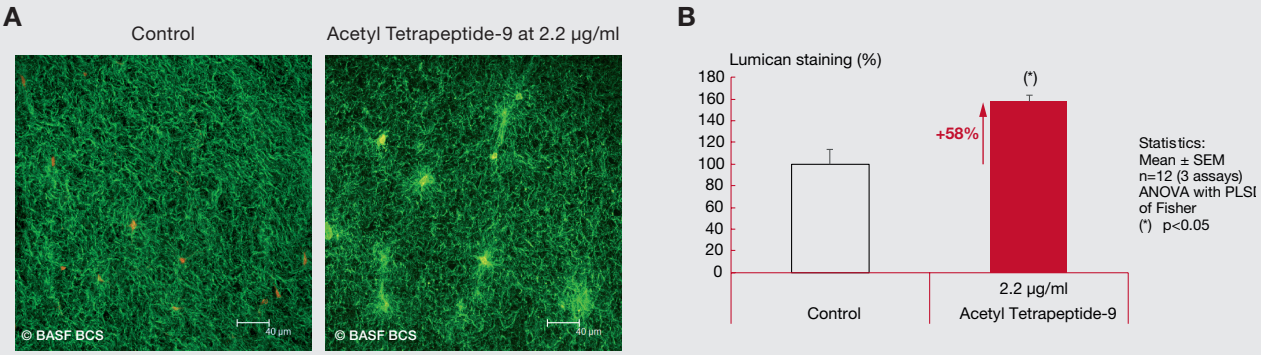


Figure 13 - Stimulation of lumican synthesis by Acetyl Tetrapeptide-9 in aged human dermal fibroblasts in pseudo-dermis. **A**: Visualization (ICC, lumican labelled in green). Scale bar: 40 µm. **B**. Quantification of staining as number of stained pixels x fluorescence intensity in green channel, expressed as a percentage of untreated control (image analysis).

CONCLUSION

Acetyl Tetrapeptide-9 in Replexium® increases the synthesis of lumican by aged human dermal fibroblasts *in vitro*, in order to improve the organization and stabilization of collagen fibril during aging.

MATERIALS&METHODS

Cell culture

Human primary dermal fibroblast from 3 donors were obtained from human skin biopsies by proteolysis. The in vitro aging of fibroblasts resulted from more than 15 repetitive passages (Hayflick model [Passi, 1997]).
For the assays in the monolayer cell culture, the cells were cultured in DMEM for 2 days at 37°C and 5% CO_2, then treated with Interleukin-4 (IL-4) at 0.1 µg/ml as positive control (Tebu, France) or Acetyl Tetrapeptide-9 and incubated 6 days at 37°C and 5% CO_2 before analysis. For the assay on pseudo-dermis, the cells were seeded in collagen I lattices with Acetyl Tetrapeptide-9 and incubated for 6 days at 37°C and 5% CO_2 before fixation and inclusion in paraffin. Sections of 7 µm were stacked on slides before analysis.

Lumican visualization and quantification

Lumican was visualized by immunohistochemistry using monoclonal antibodies (Seikagaku, Japan, ref 270427). The staining was analyzed by confocal laser scanning microscopy (Zeiss, France) and the pictures were recorded for illustration and quantification by an image analyzer analyzer as the number of stained pixels x fluorescence intensity in the green channel.

Statistics

Results are normalized to the untreated control and are expressed as the mean (%) ± SEM (3 independent assays, n=12). The statistical analysis was done using ANOVA PLSD of Fisher following SigmaPlot® software recommendations (Systat Software Inc., USA). The threshold of significance was set to 5% ($p<0.05$).

EFFICACY

Stimulation of human dermal fibroblast number and collagen I synthesis by Acetyl Tetrapeptide-9

in vitro

OBJECTIVE

The progressive atrophy of the dermis with aging is also due to the decrease of total number of fibroblasts [Gunin *et al*, 2011] and to reduced collagen I synthesis [Quan and Fisher, 2015], both contributing to an alteration of the extracellular matrix.
The aim of the study was to evaluate the effect of Acetyl Tetrapeptide-9 on the number of human dermal fibroblasts and their collagen I synthesis *in vitro*.

RESULTS & DISCUSSION

Number of human dermal fibroblast in culture

EGF, a positive reference, was used to validate the model, with measurement of cellular DNA level after 3 days of proliferation (27% increase vs control, p<0.001, data not shown).

Acetyl Tetrapeptide-9 significantly improved the proliferation of human dermal fibroblasts (Figure 14), resulting in an increased number of fibroblasts up to 10% *vs* control at 7.4 µg/ml (p<0.05).

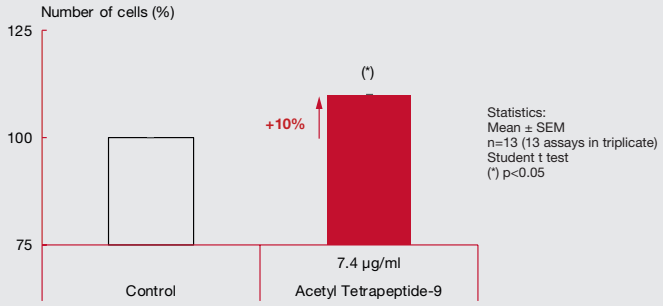


Figure 14 - Evaluation of the number of human dermal fibroblasts after 3 days of culture with Acetyl Tetrapeptide-9. Quantification of cells done through measurement of cellular DNA.

Collagen I synthesis by dermal fibroblast in culture

TGFβ1, a positive reference, was used to validate the model, with measurement of collagen I level (65% increase *vs* control, p<0.01, data not shown).

Acetyl Tetrapeptide-9 showed a significant stimulation of collagen I synthesis by human fibroblasts (Figure 15), up to 112% *vs* control at 7.4 µg/ml (p<0.05).

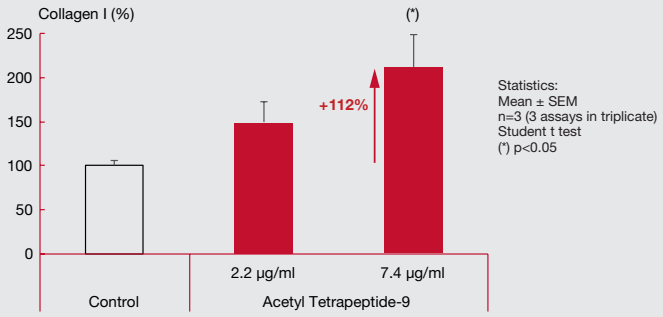


Figure 15 - Evaluation of collagen I synthesis by human dermal fibroblasts with Acetyl Tetrapeptide-9.

CONCLUSION

Acetyl Tetrapeptide-9 in Replexium® increases the number of human dermal fibroblasts and their synthesis of collagen I *in vitro* which both decrease with age, in order to contribute to the improvement of the collagen-rich connective tissue in the dermis.

MATERIALS&METHODS

Cell number assay

Human primary dermal fibroblasts were obtained from human skin biopsies by proteolysis. The cells were cultured for 1 day at 37°C and 5% CO_2, then treated with Epidermal Growth Factor (EGF) at 0.01 µg/ml or Acetyl Tetrapeptide-9 and incubated for 3 days at 37°C and 5% CO_2 before analysis of cellular DNA level using ethidium bromide.

Collagen I assay

Human primary dermal fibroblasts were obtained from human skin biopsies by proteolysis. For the assays, cells were cultured for 3 days at 37°C and 5% CO_2, then treated with TGF-β1 at 3 ng/ml as a positive control or Acetyl Tetrapeptide-9 and incubated 4 days at 37°C and 5% CO_2 before analysis of released collagen by DOT-BLOT method with monoclonal antibodies (Oncogene CP17L, France).

Statistics

Results are normalized to the untreated control and are expressed as the mean (%) ± SEM. The statistical analysis was done using Student's t test following SigmaPlot® software recommendations (Systat Software Inc. USA). The threshold of significance was set to 5% (p<0.05).

EFFICACY

Stimulation of syndecan-1 synthesis by Acetyl Tetrapeptide-11

in vitro

OBJECTIVE

Syndecan-1 proteoglycan is an important modulator of epidermal cell signaling, proliferation and cell-cell and cell-matrix adhesion [Ojeh *et al*, 2008] but decreases with age [Wegrowski *et al*, 2005; Oh *et al*, 2011].
The aim of the study was to evaluate the effect of Acetyl Tetrapeptide-11 on the synthesis of syndecan-1 by human epidermal keratinocytes in monolayer cell culture and in human full thickness reconstructed skin.

RESULTS & DISCUSSION

On monolayer keratinocytes culture

KGF, a positive reference, was used to validate the model, with measurement of syndecan-1 level by immunohistochemistry and image analysis (46% increase vs control, data not shown).

Acetyl Tetrapeptide-11 significantly stimulated syndecan-1 synthesis by aged human fibroblasts in a dose-dependent manner (Figure 16A and 16B), up to a 130% increase vs control at 2.6 µg/ml ($p<0.001$).

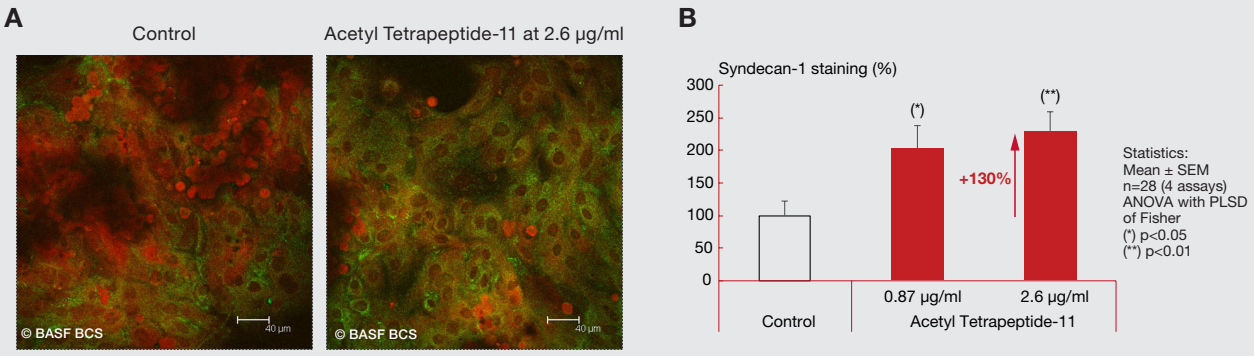


Figure 16 - Stimulation of syndecan-1 synthesis by Acetyl Tetrapeptide-11 in human epidermal keratinocytes in monolayer culture. Scale bar: 40 µm. **A**: Visualization (ICC, syndecan-1 labelled in green). **B**. Quantification of staining as number of stained pixels x fluorescence intensity in the green channel, expressed as a percentage of untreated control (image analysis).

On human reconstructed full thickness skin

Acetyl Tetrapeptide-11 at 2.6 µg/ml significantly stimulated by 38% ($p<0.05$) syndecan-1 synthesis in the epidermis of the reconstructed full thickness skin (Figure 17A and 17B).

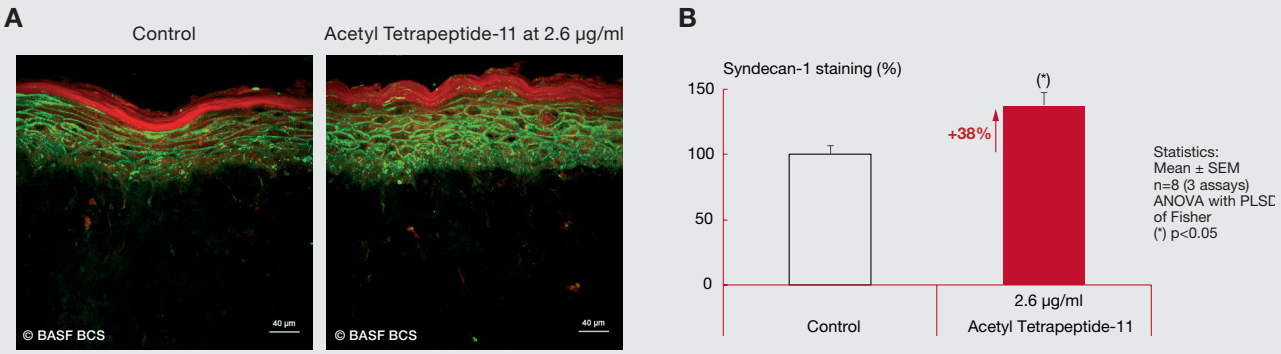


Figure 17 - Stimulation of syndecan-1 synthesis by Acetyl Tetrapeptide-11 in epidermis of reconstructed full thickness skin. Scale bar: 40 µm. **A**: Visualization (IHC, syndecan-1 labelled in green). **B**. Quantification of staining as number of stained pixels x fluorescence intensity/epidermal thickness in the green channel, expressed as a percentage of untreated control (image analysis).

CONCLUSION

Acetyl Tetrapeptide-11 increases the synthesis of syndecan-1 by human epidermal keratinocytes *in vitro* in order to improve the altered proliferation potential and cohesion of keratinocytes in an aging epidermis.

MATERIALS&METHODS

Keratinocytes monolayer cell culture

Human epidermal keratinocytes were obtained from human skin biopsies (adult donors) by trypsinization. Keratinocytes were cultured for 2 days at 37°C and 5% CO_2, then treated with Keratinocyte Growth Factor, KGF (Sigma, France), at 0.01 µg/ml as a positive control or Acetyl Tetrapeptide-11 and incubated for 5 days at 37°C and 5% CO_2 before analysis.

Reconstructed full thickness skin

Reconstructed full-thickness skins (MatTek, USA) composed of a layer of growing keratinocytes overlaid onto a pseudo-dermis of viable fibroblasts in a collagen matrix, were cultivated 1 day at 37°C and 5% CO_2, then were treated with Acetyl tetrapeptide-11 at the beginning of the differentiation phase. After 9 days of incubation at 37°C and 5% CO_2, the skins were analyzed by immunohistochemistry on sections.

Syndecan-1 visualization and quantification

Syndecan-1 was visualized by immunohistochemistry using polyclonal antibodies (SC-5632, Tebu, France), the staining was analyzed by confocal laser scanning microscopy (Zeiss, France) and the pictures were recorded for illustration and quantification by an image analyzer as the number of stained pixels x fluorescence intensity in the green channel/epidermis thickness.

Statistics

Results are normalized to the untreated control and are expressed as the mean (%) ± SEM. The statistical analysis was done using ANOVA PLSD of Fisher following SigmaPlot® software recommendations (Systat Software Inc. USA). The threshold of significance was set to 5% ($p<0.05$).

EFFICACY

Stimulation of human epidermal keratinocyte proliferation by Acetyl Tetrapeptide-11

in vitro

OBJECTIVE

Skin aging is characterized by a decrease of epidermal cell proliferation in the basal layer [Yaar *et al*, 2002].
As syndecan-1 is an important modulator of epidermal cell proliferation, the aim of the study was to evaluate the stimulatory effect of Acetyl Tetrapeptide-11 on the proliferation of human epidermal keratinocytes.

RESULTS & DISCUSSION

Foetal Calf Serum (FCS), a positive reference, was used to validate the model, with measurement of cellular DNA level after 3 days of proliferation (18% increase vs control, data not shown).

Acetyl Tetrapeptide-11 significantly improved proliferation of human epidermal keratinocytes in a dose-dependent manner (Figure 18), resulting in an increase in the number of keratinocytes up to 13% vs control at 0.87 µg/ml ($p<0.05$).

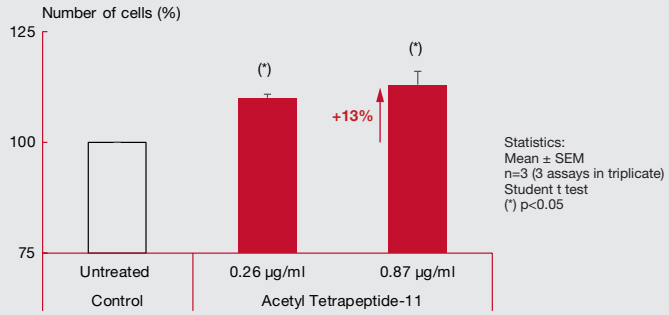


Figure 18 - Cellular growth of human epidermal keratinocytes after 3 days of culture. Quantification of cells done through measurement of cellular DNA.

CONCLUSION

In vitro, Acetyl Tetrapeptide-11 increases the proliferation of human epidermal keratinocytes which decreases with age, in order to contribute to the improvement of the epidermal homeostasis and cohesion.

MATERIALS&METHODS

Cell proliferation assay

Human primary epidermal keratinocytes were obtained from human skin biopsies by proteolysis. The cells were cultured for 1-2 days at 37°C and 5% CO_2, then treated with FCS at 10 mg/ml as a positive control or Acetyl Tetrapeptide-11 and incubated for 3 days at 37°C and 5% CO_2 before analysis of cellular DNA level using ethidium bromide.

Statistics

Results are normalized to the untreated control and are expressed as the mean (%) ± SEM. The statistical analysis was done using Student's t test following SigmaPlot® software recommendations (Systat Software Inc. USA). The threshold of significance was set to 5% ($p<0.05$).

EFFICACY

Stimulation of collagen XVII synthesis by Acetyl Tetrapeptide-11

in vitro

OBJECTIVE

Collagen XVII is a transmembrane protein in the DEJ, anchoring epidermal basal cells to the basement membrane by interacting with both intracellular and extracellular components [Powell *et al*, 2005]. But Collagen XVII decreases with age, contributing to the loss of structural integrity of the aging DEJ and its increased fragility.
The aim of the study was to evaluate the effect of Acetyl Tetrapeptide-11 on collagen XVII by human epidermal keratinocytes.

RESULTS & DISCUSSION

Acetyl Tetrapeptide-11 significantly stimulated collagen XVII synthesis by human epidermal keratinocytes in a dose-dependent manner (Figure 19A and 19B), up to 185% increase *vs* control at 2.6 µg/ml ($p<0.01$).

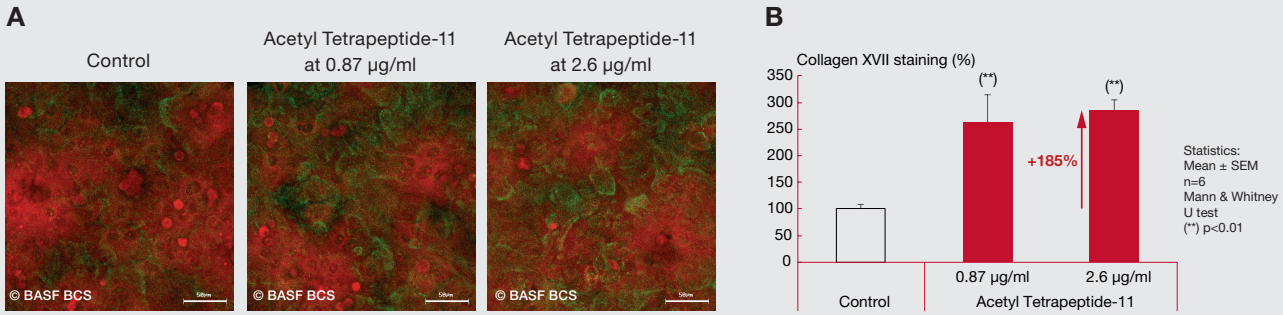


Figure 19 - Stimulation of collagen XVII synthesis by Acetyl Tetrapeptide-11 in human epidermal keratinocytes. Scale bar: 40 µm. **A**: Visualization (ICC, collagen XVII labelled in green). **B**. Quantification of staining as percentage of coverage of the cell culture, expressed as percentage of the untreated control (image analysis).

CONCLUSION

Acetyl Tetrapeptide-11 in Replexium® increases the synthesis of collagen XVII by human epidermal keratinocytes in vitro, in order to improve the structural integrity of the DEJ and the cohesion between the epidermis and dermis altered by aging.

MATERIALS&METHODS

Cell culture

Human keratinocytes were isolated by trypsinization from human skin biopsies. They were cultured in a standard medium containing 10% FCS (fetal calf serum), then incubated at 37°C in the presence of 5% CO_2.
After 2 days, the culture medium was changed and replaced by a medium containing Acetyl Tetrapeptide-11. Human keratinocytes were incubated for 3 days at 37°C in the presence of 5% CO_2 before analysis.

Collagen XVII visualization and quantification

Collagen XVII was visualized by immunohistochemistry using a monoclonal antibody (Abcam, France), the staining was analyzed by confocal laser scanning microscopy (Leica, France) and the pictures were recorded for illustration and quantification by an image analyzer as the surface covered by immunostaining.

Statistics

Results are normalized to the untreated control and are expressed as the mean (%) ± SEM. The statistical analysis was done using Mann & Whitney U test following SigmaPlot® software recommendations (Systat Software Inc. USA). The threshold of significance was set to 5% ($p<0.05$).

EFFICACY

Improvement of the extracellular matrix network by the complex of Acetyl Tetrapeptide-9 and Acetyl Tetrapeptide-11

in vitro

OBJECTIVE

The aim of the study was to evaluate the effect on extracellular matrix network (ECM) of the complex of the two peptides Acetyl Tetrapeptide-9 and Acetyl Tetrapeptide-11, on a 3D bioprinted skin [Cadau *et al*, 2017].
Fully printed 3D skin by Laser Assisted Bioprinting combined with 2nd harmonic generation (SHG) are advanced methods allowing to study the extracellular matrix [Pain *et al*, 2017]. It provides structural and non-invasive information on the collagen network in the dermis without exogenic staining.

RESULTS & DISCUSSION

The complex of the two peptides visibly improved the organisation of the collagen fibers network and the structure of the dermis (Figure 20).

In the dermis treated with the complex of the two peptides, we observed in its upper part a network of well-defined and very intense collagen fibers (yellow arrow in Figure 18). Comparatively, in the untreated control condition, the fibers are smaller and not well-defined, with a blurry network (red arrow in Figure 20). A similar improvement of the structure and organization of the collagen network, with some collagen fibers organized in clusters, was observed deeper in the bioprinted dermis (data not shown).

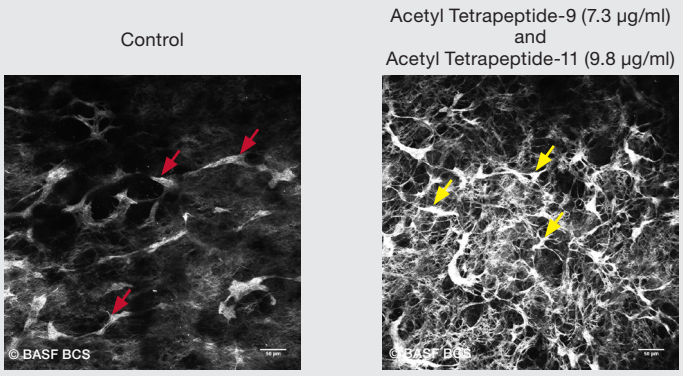


Figure 20 - Visualization by second harmonic generation (SHG) of the effect of Acetyl Tetrapeptide-9 and Acetyl Tetrapeptide-11 complex on the collagenic network in 3D bioprinted skin. Collagen fibers are visualized in grey. Images a 3D stack acquisition in depth of 10 planes of the upper part of the dermis. Scale bar: 50 µm.

CONCLUSION

Replexium® peptide complex improves the density and organization of the extracellular collagen fibers in the dermis of 3D bioprinted skin, for a denser and firmer skin.

MATERIALS&METHODS

3D bioprinted skin

Reconstructed full bioprinted skins are alternatively composed of multi-layers of rat collagen type I and III and of human fibroblasts using a microvalve and laser technology respectively. Bioprinted dermis were matured for 5 days in DMEM/F12 medium with 10% bovine calf serum, 100 µg/ml normocine, 5 µg/ml ascorbic acid. Human keratinocytes were printed on top of the dermis and matured for 10 days in modified Green mediums 0.8% BSA, 0.12 UI/ml insulin, 0.4 µg/ml hydrocortisone and 5µg/ml ascorbic acid. The cultures were made at 37°C and 5% CO_2.

Treatments with a solution of Acetyl Tetrapeptide-9 at 7.3 µg/ml and Acetyl Tetrapeptide-11 at 9.8 µg/ml were done daily except when cells were seeded.

2nd harmonic generation (SHG) microscopy

We used confocal microscopy (LSM 780, Zeiss, France) associated with a multiphoton laser (Spectra Physics) to generate a second harmonic signal specific from fibrillar collagen. The acquisition is done with whole bioprinted skin without histological cuts. The images were acquired in depth from epidermis to dermis to create a 3D stack from 10 planes. This stack of images is used for 2D visualization of the effect of the treatment, comparatively with the untreated control condition.

EFFICACY

Appearance of wrinkle reduction and skin firming effects of Replexium®

in vivo

OBJECTIVE

In a double blind clinical study, we evaluated the effects of Replexium® for its ability to improve the appearance of signs of skin aging, wrinkles on the face and dermal density. This study was done with Replexium® at 2% compared to a leading Peptide-blend benchmark at its recommended dose of use (3%) (Figure 21).

Aging causes structural changes in the skin, in particular in the extracellular matrix, epidermal cohesion and anchoring between the epidermis and the dermis at the DEJ level, leading to the loss of dermal density and firmness, and to the appearance of wrinkles on the face.

Caucasian females, ages 45-65 years, having visible forehead wrinkles, crow's feet wrinkles, nasolabial folds, low dermal density and considering themselves to have loss of skin elasticity and firmness.
Double-blind, randomized split face, twice daily applications for 56 day

Replexium® BC10091 at 2%



Peptide-blend benchmark at 3%

Grading of forehead wrinkles, crow's feet wrinkles and nasolabial folds on Visia Images.
Measurement of dermal density and of hypoechogenic dermal factor by ultrasound on the cheek area.
Time points: D0, D21 and D56.

Figure 21 - Clinical study on wrinkles and dermal density of Replexium® vs a Peptide-blend benchmark.

RESULTS & DISCUSSION

Wrinkles and nasolabial folds appearance improvement

Replexium® at 2% demonstrated a significant improvement of forehead wrinkles, crow's feet wrinkles and nasolabial folds as graded by a clinical scientist (Figure 22A, 22B, 22C). After 21 days of use, forehead and crow's feet wrinkle grades were both improved by 11% ($p<0.001$) and nasolabial folds were improved by 9% ($p<0.001$) vs baseline. This anti-wrinkle effect improved over time, with clinical grades for crow's feet and forehead wrinkles improved at D56 by 23% ($p<0.001$) and for nasolabial folds at D56 by 19% ($p<0.001$).

These results with Replexium® at 2% are similar to the results obtained with the Peptide-blend benchmark at 3%, without any significant differences between the 2 products (Figure 22A, 22B, 22C).

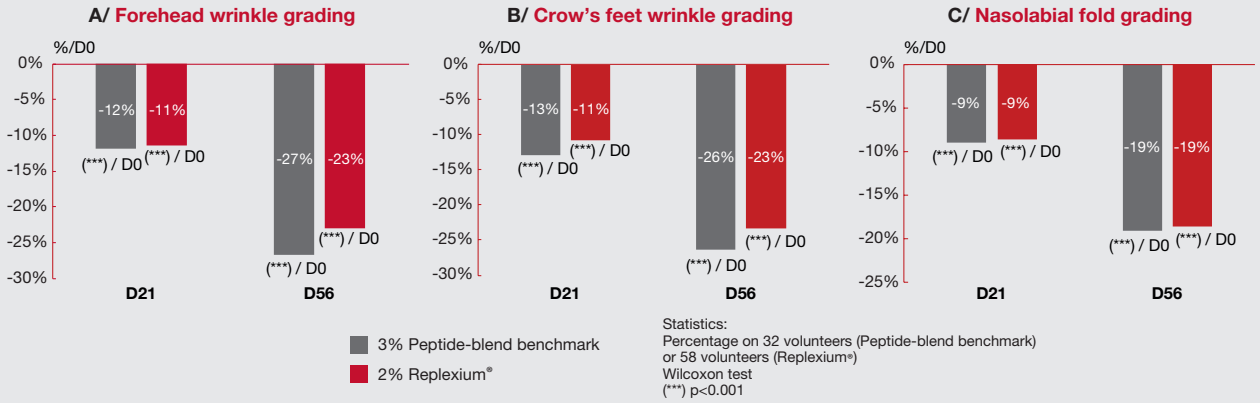


Figure 22 - Percentage of evolution of clinical grading results *vs* baseline of A/ Forehead wrinkles, B/ Crow's feet wrinkles and C/ Nasolabial folds for Replexium® at 2% and Peptide-blend benchmark at 3% at D21 and D56.

in vivo

Pictures in figure 23 illustrate the improvement of wrinkles and nasolabial folds with Replexium®. The effect is visible after 21 days and improved after 56 days.

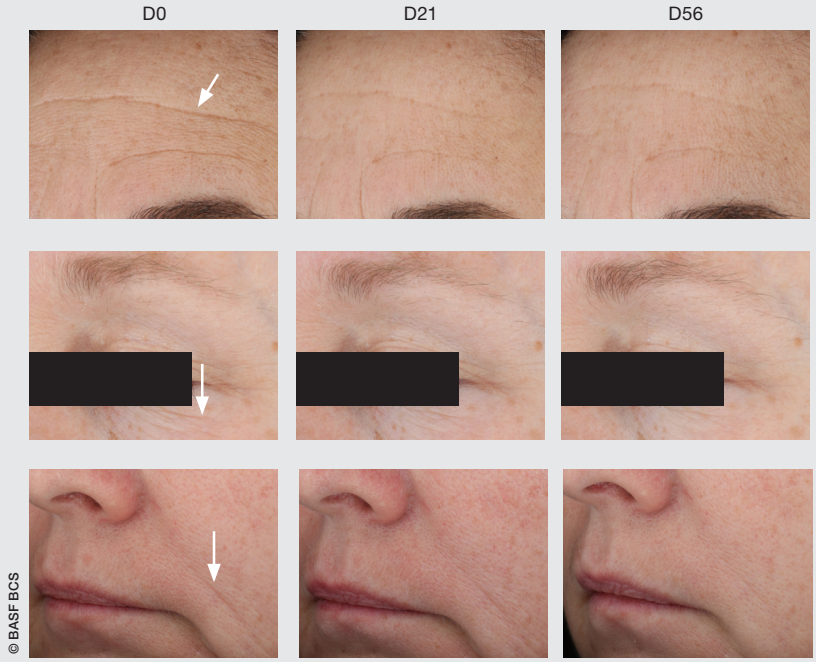


Figure 23 - Illustrative pictures of the effect of Replexium® at 2% on the signs of aging. D0: before treatment, D21 and D56: after 21 and 56 days of product application.

Dermal density and hypoechogenic dermal factors for skin firmness improvement

The dermal density and the hypoechogenic dermal factor measured by high-frequency ultrasound are factors reflecting the structure of the dermis and are indicators of skin aging [Crisan *et al*, 2012]. Hypoechogenic density factor is the percentage of low echogenic pixels (intensity interval: 0-30) and increases with age. An improvement of the collagen I fibril structure in the extracellular matrix should result in an increase of dermal density and a decrease of hypoechogenic dermal factor, meaning an improvement of the density of the dermis, and consequently of the skin firmness.

After 21 days, Replexium® demonstrated a significant 21% increase of dermal density value ($p<0.001$) and a significant decrease by 15% of the hypoechogenic dermal factor ($p<0.001$) on the face versus before treatment (D0), whereas in the same time no change was observed with the Peptide-blend benchmark (Figure 24A and 24B). These improvements of dermal density value and of hypoechogenic dermal factor with Replexium® were significant *vs* the Peptide-blend benchmark (respectively 21% and 17%; $p<0.001$), demonstrating the quicker skin firming effect of Replexium®.

With the Peptide-blend benchmark, 56 days of use were needed to observe some significant changes of the dermal density value and hypoechogenic dermal factor (Figure 24A and 24B). At the same time, at D56, the significant changes observed at D21 with Replexium® were confirmed. Replexium® overperformed the Peptide-blend benchmark (even though the differences between the 2 products are not significant at D56).

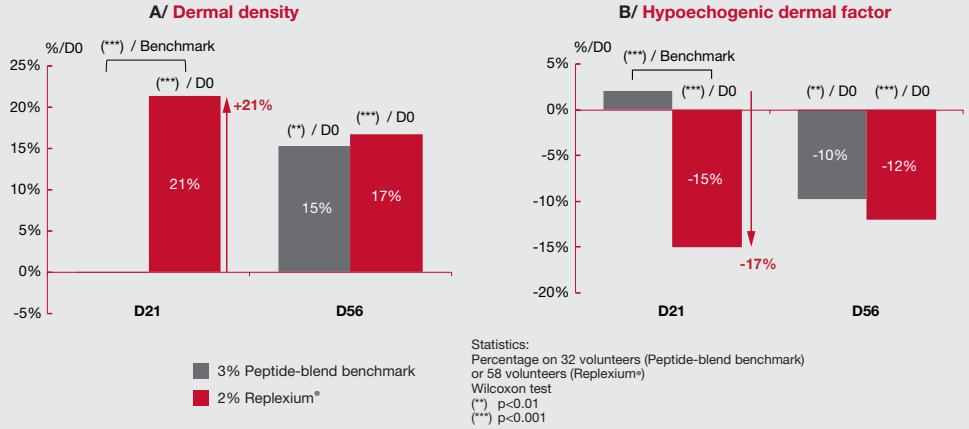


Figure 24 - Percentage of evolution of **A**: dermal density, and **B**: hypoechogenic dermal factor vs baseline measured by ultrasounds on the face for Replexium® at 2% and the Peptide-blend benchmark at 3% at D21 and D56.

in vivo

The ultrasound images in figure 25 illustrate the changes over time of the dermal density (green pixels) and hypoechogenic density factor (low echogenic pixels with black color) with the Peptide-blend benchmark and Replexium® on a volunteer.
We can observe more green pixels and less low echogenic black pixels with Replexium® (arrows) at D21 whereas no changes are visible with the peptide-blend benchmark at the same time.

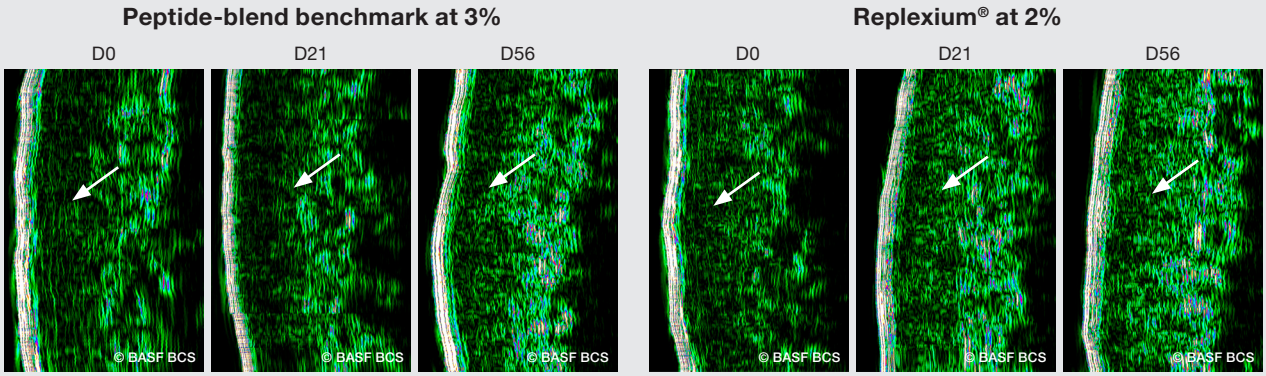


Figure 25 - Illustrative echography pictures of the dermal density (green pixels) and hypoechogenic density factor (low echogenic pixels with black color) change over time (Baseline (D0) and after 21 (D21) and 56 (D56) days of product use for the Peptide-blend benchmark at 3%, and Replexium® at 2%).

Estimation of age gain corresponding to the change of the dermal density at 21 days

The age gain corresponding to the increase of dermal density was estimated from the difference observed in dermal density between two age groups of volunteers.

The dermal density was measured with ultrasound on the face of 24 healthy women distributed into 2 age groups: group I, n=12, 55-60 years with average age 57.4, and group II, n=12, 61-65 with average age 63.0. On each volunteer, two measurements were done on the right and left half-faces, and all the measurements were pooled according the age groups. There was a significant ($p<0.05$) decrease of the dermal density on the face, with a difference of 6.0 AU between the average values of the two groups (Figure 26).

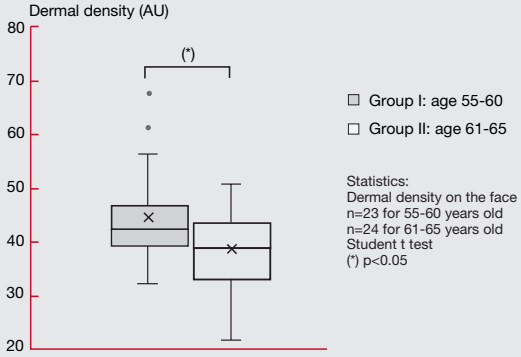


Figure 26 - Dermal density of the face measured by ultrasound arranged by age group.

By extrapolation, based on this evaluation of influence of age on the dermal density, the improvement observed with Replexium® at D21 would be comparable to a flashback to 7 years of younger and firmer skin.

CONCLUSION

Replexium® visibly improves wrinkles (forehead and crow's feet wrinkles, nasolabial fold) and skin firmness with an effect on the density of dermal structure.

This firming effect with Replexium® is quicker than with the Peptide-blend benchmark. After 21 days, the density of dermal structure increases whereas no effect is observed with the Peptide-blend benchmark. Based on an evaluation of the influence of age on the dermal density, this effect would be comparable to gaining 7 years of younger and firmer skin.

The anti-wrinkle effect of Replexium® is similar to the effect of the Peptide-blend benchmark. Wrinkle and nasolabial fold appearance is reduced as early as 21 days of product use, and improves over time.

MATERIALS&METHODS

Anti-aging study

Study design
The clinical trial was carried out as a double-blind randomized study, under dermatological control. The efficacy of the formulation containing Replexium® BC10091 at 2% (corresponding to Acetyl Tetrapeptide-11 at ~675 ppm, Acetyl Tetrapeptide-9 at ~500 ppm, Dimethyl Isosorbide at 0.8% and Polysorbate 20 at 0.8%) compared to the same formula containing the peptide-blend benchmark at its recommended dose of use of 3%. The quantity of bioavailability enhancers Dimethyl Isosorbide and Polysorbate 20 was adjusted to have the same content in the two formulations. The formulations are detailed in Annex 2. The study was conducted during a period of 56 days with check points at D0, D21 and D56.

Inclusion criteria
The study was done on healthy Caucasian female volunteers, Fitzpatrick skin type I, II or III, age 45 to 65, considering themselves to have loss of skin elasticity and firmness, having visible crow's feet wrinkles of grade 3-4 according to the Bazin Atlas (Bazin, 2007), forehead wrinkles, and nasolabial folds of grade 2-3 according to the Bazin Atlas, assessed by a clinical scientist, and low dermal density (dermal density ≤ 45 measured by ultrasound).

Application modality
58 volunteers applied the Replexium® formula on half of the face and 32 volunteers applied the Peptide-blend benchmark formula on half the face. The formulations were applied twice a day for 56 days, under normal conditions of use.

Evaluation methods
Dermal density with ultrasound
The DUB SkinScanner is a high-resolution diagnostic ultrasound system with a 22MHz Probe. It is used to evaluate dermal density. The system works by sending high-frequency ultrasound impulses into the skin. The ultrasound waves reflected by the structure within the skin are digitally processed and displayed as an image. The high-frequency ultrasound and its software allow the assessment and analysis of skin dermal density and hypoechogenic dermal factor, which are indicators of skin aging.
Wrinkles and nasolabial folds grading
The evaluation was done by 4 trained clinical scientists from facial images taken from the VISIA-CR 2D Imaging System. The grading of forehead wrinkles and nasolabial folds was done with a 0-5 scale with 0.5 intervals and the grading of the crow's feet wrinkles was done with a 0-6 scale with 0.5 intervals according to the Bazin Atlas (Bazin, 2007).

Statistics
Results are expressed as the mean percentage of change compared to the baseline measurement (D0).
The statistical analysis of the parameters has been done after the verification of the normality of the distribution using Shapiro-Wilk test. Afterwards, the statistical analysis of the change in the studied parameters over time was performed with paired Student t test if the normality of distribution had been confirmed, or with Wilcoxon test in the opposite case. The statistical analysis of the differences in the studied parameters between the treatment groups was performed with unpaired Student t test if the normality of distribution had been confirmed, or with Mann-Whitney test in the opposite case.

Influence of age on the dermal density

Subjects
24 healthy women, Fitzpatrick skin type I, II or III, considering themselves to have loss of skin elasticity and firmness were included in the analysis. They were distributed into 2 age groups: group I, n=12, 55-60 years with average age 57.4, and group II, n=12, 61-65 with average age 63.0.

Measurements of dermal density
Measurements were done after a washout period of 14 days and after 30 minutes of acclimation in the laboratory. The DUB SkinScanner ultrasound system with a 22 MHz Probe was used to measure dermal density. On each volunteer, a measurement was done on the right and left half-face.

Statistics
All measurements were pooled according the age groups. The statistical analysis of the parameters has been done after the verification of the normality of the distribution using Shapiro-Wilk test. The statistical analysis of the difference in the studied parameters between the two age groups was performed with unpaired Student t test if the normality of distribution had been confirmed, or with Mann-Whitney test in the opposite case.



GENERAL CONCLUSION

Achieve your beauty aspiration

Replexium® is a skin-bioavailable synergistic complex of 2 patented peptides for the development of performance skin care products. Replexium® demonstrates quick efficacy by reducing the appearance of wrinkles and provides skin firming benefits within 3 weeks – faster than peptide benchmark.

Women aspire to look 5-10 years younger than their actual age, and expect a quality skin-care product would help them achieve their aspirations.

During aging, various complex structural and functional changes occur in the skin, with a progressive atrophy of the dermis and epidermis. Not only is the dermis impacted, with loss and disorganization of collagen fibrils in the extracellular matrix, but so is the epidermis, with a reduced mitosis rate and a loss of cellular cohesion, and as well as the dermal-epidermal junction (DEJ) with a flattening and a less effective anchoring system.

To improve the signs of skin aging, we developed Replexium®, a synergistic complex combining Acetyl Tetrapeptide-9 and Acetyl Tetrapeptide-11 with bioavailability agents, targeting extracellular matrix, epidermal, and DEJ structural integrity.

The Acetyl Tetrapeptide-9 in Replexium® increased the age-affected synthesis of lumican proteoglycan and the number of dermal fibroblasts *in vitro*, to improve the impaired collagen fibril organization in the aging dermis. The Acetyl Tetrapeptide-11 in Replexium® increased the age-affected synthesis of syndecan-1 proteoglycan and of collagen XVII, and the number of epidermal keratinocytes *in vitro*, to improve the proliferation potential, cohesion and anchoring of keratinocytes in the aging epidermis and DEJ.

On 3D bioprinted skin, the complex of the 2 tetrapeptides improved the density and organization of the extracellular collagen fibers for a denser dermis.

Tested *in vivo* in a clinical test on human volunteers compared to 3% of a Peptide-blend benchmark, Replexium® at 2% demonstrated a quicker effect on dermal density and similar anti-wrinkle performance. Replexium® visibly improved the appearance of wrinkles and the density of the dermal structure, for a fast anti-wrinkle and skin firming efficacy within 3 weeks. Based on an internal study on dermal density, we evaluated that this improvement is comparable to a 7 years firmer skin.

Meet the consumer expectations with Replexium®: gain 7 years firmer skin in 21 days with visible wrinkle reduction.

In vitro studies

+ 115%	Lumican level (dermal fibroblasts)
+ 112%	Collagen I level (dermal fibroblasts)
+ 10%	Fibroblasts cell number
+ 130%	Syndecan-1 level (epidermal keratinocytes)
+ 13%	Keratinocytes cell number
+ 185%	Collagen XVII level (epidermal keratinocytes)

Skin models studies

+ 58%	Lumican level in pseudo dermis
+ 38%	Syndecan-1 level in reconstructed skin
↗ visibly	Fibers in 3D bioprinted skin

In vivo study

- 11%	Wrinkle reduction at day 21
- 23%	Wrinkle reduction at day 56
+ 21%	Skin firming at day 21 (faster than Peptide-blend benchmark) (comparable to gaining 7 years firmer skin)

ANNEXES

Annex 1 - Technical data - Available upon request

- Quality and Regulatory Product Information
- Composition sheet
- Specifications
- Formulation Data Sheet

Annex 2 - Clinical test formula

Trade name	INCI name	Peptide-blend benchmark formulation %	Replexium® formulation %
Deionized Water	Water	sqf 100	sqf 100
Gransolve DMI (Grant Industries)	Dimethyl Isosorbide	2.00[b]	1.20[b]
Eumulgin® SG	Sodium Stearoyl Glutamate	0.08	0.08
Glycerin	Glycerin	2.00	2.00
Keltrol CG-T (CP Kelco)	Xanthan Gum	0.10	0.10
Cosmedia® SP	Sodium Polyacrylate	1.50	1.50
Plantapon® LGC-Sorb	Sodium Lauryl Glucose Carboxylate (and) Lauryl Glucoside	1.75	1.75
Dehymuls® PGPH	Polyglyceryl-2 Dipolyhydroxystearate	2.00	2.00
Cetiol® C5	Coco-Caprylate	6.50	6.50
Eumulgin® SML-20	Polysorbate 20	2.00[b]	1.20[b]
Euxyl PE 9010 (Schülke)	Phenoxyethanol (and) Ethylhexylglycerin	1.10	1.10
Peptide-blend benchmark	-	3.00	
Replexium® BC10091[a]	Dimethyl Isosorbide (and) Polysorbate 20 (and) Water (and) Acetyl Tetrapeptide-11 (and) Acetyl Tetrapeptide-9	-	2.00[a]

[a] Corresponds to Acetyl Tetrapeptide-11 at ~675 ppm, Acetyl Tetrapeptide-9 at ~500 ppm, Dimethyl Isosorbide at 0.80% and Polysorbate 20 at 0.80%
[b] The quantity of Dimethyl Isosorbide and Polysorbate 20 was adjusted to have exactly the same content in the two formulations: 2.00% for each component.

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Annex 3 - Formulation examples

Miracle day cream (SC-FR-18-BC-50839-01)

Phase	Ingredients	INCI	% by weight	Function
A	Eumulgin® VL 75	Lauryl Glucoside, Polyglyceryl-2 Dipolyhydroxystearate, Glycerin	1.50	Emulsifier (O/W)
	Lanette® 22	Behenyl Alcohol	1.00	Consistency agent
	Cutina® PES	Pentaerythrityl Distearate	1.00	Consistency agent
	Cetiol® SB 45	Butyrospermum Parkii Butter	2.00	Emollient
	Cetiol® C 5C	Coco-Caprylate/Caprate	5.00	Emollient
	Cetiol® Sensoft	Propylheptyl Caprylate	4.00	Emollient
	Cegesoft® PS 6	Olus Oil [EU], Vegetable Oil [CTFA]	2.00	Emollient
	Dow Corning 9041 Silicone Elastomer Blend *(Dow Corning)*	Dimethicone, Dimethicone Crosspolymer	2.00	Skin feel modifier
	Cosmedia® SP	Sodium Polyacrylate	0.60	Rheology modifier
B	Water, demin.	Aqua	74.90	
	Glycerin	Glycerin	3.00	Humectant
	Preservative*		qs	Preservative
C	Dry-Flo Plus *(Akzo Nobel)*	Aluminum Starch Octenylsuccinate	1.00	Skin feel modifier
D	Replexium™ BC10091	Aqua, Dimethyl Isosorbide, Polysorbate 20, Acetyl Tetrapeptide-11, Acetyl Tetrapeptide-9	2.00	Active ingredient
	Perfume*	Parfum	qs	Fragrance

Specifications	
pH value (23°C)	6.30
Viscosity (Brookfield; RVT; spindle TC, Helipath; 20 rpm; 23°C)	68 000 mPas

SWOP powered clinical formula (SC-FR-18-BC-50846-02)

Phase	Ingredients	INCI	% by weight	Function
A	Water, demin.	Aqua	82.97	
	Euxyl PE 9010 *(Schülke)*	Phenoxyethanol, Ethylhexylglycerin	1.10	Preservative
	Replexium™ BC10091	Aqua, Dimethyl Isosorbide, Polysorbate 20, Acetyl Tetrapeptide-11, Acetyl Tetrapeptide-9	2.00	Active ingredient
B	Eumulgin® SG	Sodium Stearoyl Glutamate	0.08	Emulsifier (O/W)
C	Glycerin	Glycerin	2.00	Humectant
	Rheocare® XGN	Xanthan Gum	0.10	Stabilizer
	Cosmedia® SP	Sodium Polyacrylate	1.50	Rheology modifier
D	Plantapon® LGC SORB	Sodium Lauryl Glucose Carboxylate (and) Lauryl Glucoside	1.75	Emulsifier (O/W)
E	Dehymuls® PGPH	Polyglyceryl-2 Dipolyhydroxystearate	2.00	Emulsifier (W/O)
	Cetiol® C 5C	Coco-Caprylate/Caprate	6.50	Emollient
F	Perfume*	Parfum	q.s.	Fragrance

Specifications	
pH value (23°C)	5.95
Viscosity (Brookfield; RVT; spindle TC, Helipath; 5 rpm; 23°C)	16 000 mPas

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Rapid Wrinkle Repair Firming Serum (SC-FR-18-BC-50837-01)

Phase	Ingredients	INCI	% by weight	Function
A	Eumulgin® SG	Sodium Stearoyl Glutamate	0.05	Emulsifier (O/W)
	Lanette® O	Cetearyl Alcohol	2.00	Consistency agent
	Cetiol® C 5C	Coco-Caprylate/Caprate	2.50	Emollient
	Cetiol® CC	Dicaprylyl Carbonate	1.50	Emollient
	Cetiol® Sensoft	Propylheptyl Caprylate	1.50	Emollient
	Myritol® 312	Caprylic/Capric Triglyceride	2.00	Emollient
B	Water, demin.	Aqua	50.43	
	Glycerin	Glycerin	3.00	Humectant
C	Water, demin.	Aqua	33.00	
	Rheocare® C Plus	Carbomer	0.20	Rheology modifier
D	Euxyl PE 9010 *(Schülke)*	Phenoxyethanol, Ethylhexylglycerin	1.10	Preservative
	Sensiva SC 50 *(Schülke)*	Ethylhexylglycerin	0.40	Auxiliary
E	Sodium Hydroxide (18% solution)	Sodium Hydroxide	0.32	pH Adjustment
F	Replexium™ BC10091	Aqua, Dimethyl Isosorbide, Polysorbate 20, Acetyl Tetrapeptide-11, Acetyl Tetrapeptide-9	2.00	Active ingredient
	Perfume*	Parfum	qs	Fragrance

Specifications	
pH value (23°C)	6.25
Viscosity (Brookfield; RVT; spindle TC, Helipath; 20 rpm; 23°C)	35 000 mPas

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Publisher
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Global Edition, March 2018

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